

Two limit rules

Basic skills

Q1. An improper integral is one that is:

- undefined at a point in the limit
- or
- Infinity limit

Q2. $\int_0^1 \frac{1}{\sqrt{x}} dx$ undefined at $x=0$

$$= \int_0^1 x^{-\frac{1}{2}} dx = \lim_{k \rightarrow 0} \int_k^1 x^{-\frac{1}{2}} dx$$

$$= \lim_{k \rightarrow 0} [2\sqrt{x}]_k^1 = \lim_{k \rightarrow 0} (2\sqrt{1} - 2\sqrt{k}) = 2$$

Q3. a) $\lim_{x \rightarrow \infty} x^2 e^{-x} = 0$ $\leftarrow \lim_{x \rightarrow \infty} x^k e^{-x} = 0$
here $k=2$

b) $\lim_{x \rightarrow 0} x^5 \ln x = 0$ $\leftarrow \lim_{x \rightarrow 0} x^k \ln x = 0$
 $k=5$

c) $\lim_{x \rightarrow \infty} \frac{3x^5}{e^x} = 3 \lim_{x \rightarrow \infty} x^5 e^{-x} = 3 \times 0 = 0$

d) $\lim_{x \rightarrow 0} \frac{\ln x}{2\sqrt{x}} = \frac{1}{2} \lim_{x \rightarrow 0} x^{-\frac{1}{2}} \ln x = \frac{1}{2} \times 0 = 0$

Q4. $y = 2x$ so $x = \frac{y}{2}$

$$\lim_{x \rightarrow \infty} 2x^2 \sqrt{x} e^{-2x} = \lim_{x \rightarrow \infty} 2x^{\frac{5}{2}} e^{-2x}$$

$$= \lim_{y \rightarrow \infty} 2 \left(\frac{y}{2}\right)^{\frac{5}{2}} e^{-y} = \frac{2}{2^{\frac{5}{2}}} \lim_{y \rightarrow \infty} y^{\frac{5}{2}} e^{-y}$$

$$= \frac{1}{2^{\frac{3}{2}}} \times 0 = 0$$

Competency skills

a) $\lim_{x \rightarrow \infty} x^k e^{-x} = \lim_{x \rightarrow \infty} \frac{x^k}{e^x}$

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$$= \lim_{x \rightarrow \infty} \frac{x^k}{1 + x + \dots + \frac{x^k}{k!} + \dots} = \lim_{x \rightarrow \infty} \frac{1}{\frac{1}{x^k} + \frac{1}{x^{k-1}} + \dots + \frac{1}{k!} + \frac{x}{(k+1)!} + \dots}$$

$$= 0$$

b) $\lim_{x \rightarrow \infty} x^k e^{-x} = \lim_{x \rightarrow \infty} \frac{x^k}{e^x} = \lim_{x \rightarrow \infty} \frac{kx^{k-1}}{e^x}$

$$= \dots = \lim_{x \rightarrow \infty} \frac{k(k-1)\dots 1 x^0}{e^x} = 0$$

apply L'Hopital's k times in total

Q6. $x = e^{-\frac{y}{k}}$ as $x \rightarrow 0, y \rightarrow \infty$

So $\lim_{x \rightarrow 0} x^k \ln x = \lim_{y \rightarrow \infty} \left(e^{-\frac{y}{k}}\right)^k \ln \left(e^{-\frac{y}{k}}\right)$

$$= \lim_{y \rightarrow \infty} e^{-y} \times -\frac{y}{k} = -\frac{1}{k} \lim_{y \rightarrow \infty} y e^{-y}$$

$$= -\frac{1}{k} \times 0 = 0$$

$k \neq 0$

Q7. $\lim_{x \rightarrow s} (x-s) \ln(x-s) = \lim_{y \rightarrow 0} y \ln y = 0$

$y = x-s$ so $x \rightarrow s, y \rightarrow 0$

Q8. a) $\int_0^{\infty} x e^{-x} dx$ Parts

$u = x \quad v = -e^{-x}$
 $u' = 1 \quad v' = e^{-x}$

$$= \lim_{k \rightarrow \infty} \left([-x e^{-x}]_0^k + \int_0^k e^{-x} dx \right)$$

$$= \lim_{k \rightarrow \infty} [-x e^{-x} - e^{-x}]_0^k = \lim_{k \rightarrow \infty} (-\underbrace{k e^{-k}}_{\text{limit} = 0} - \underbrace{e^{-k}}_{\text{limit} = 0} - (-1))$$

$$= 0 - 0 + 1 = 1$$

$$b) \int_0^{\infty} x e^x dx$$

EXAM PAPERS PRACTICE

Parts $u=x$ $v=e^x$
 $u'=1$ $v'=e^x$

$$= \lim_{k \rightarrow \infty} \left[x e^x - e^x \right]_0^k = \lim_{k \rightarrow \infty} (\underbrace{k e^k}_{\text{limit} = \infty} - \underbrace{e^k}_{\text{limit} = \infty} - (-1))$$

$$= \infty$$

$$c) \int_0^5 x^3 \ln x dx$$

Parts $u=\ln x$ $v=\frac{1}{4}x^4$
 $u'=\frac{1}{x}$ $v'=x^3$

$$= \lim_{k \rightarrow 0} \left(\left[\frac{1}{4} x^4 \ln x \right]_k^5 - \int_k^5 \frac{1}{4} x^3 dx \right)$$

$$= \lim_{k \rightarrow 0} \left[\frac{1}{4} x^4 \ln x - \frac{1}{16} x^4 \right]_k^5$$

$$= \lim_{k \rightarrow 0} \left(\frac{1}{4} 5^4 \ln 5 - \frac{1}{16} 5^4 - \left(\underbrace{\frac{1}{4} k^4 \ln k}_{\text{limit} = 0} - \underbrace{\frac{1}{16} k^4}_{\text{limit} = 0} \right) \right)$$

$$= \frac{5^4}{4} \ln 5 - \frac{5^4}{16}$$

Advanced skills

$$Q9. \lim_{x \rightarrow 0} \left(1 - \frac{a}{(1+x)^3} \right) \ln x = \lim_{x \rightarrow 0} \left(\frac{(1+x)^3 - a}{(1+x)^3} \right) \ln x$$

$$= \lim_{x \rightarrow 0} \left(\frac{(1-a) + 3x + 3x^2 + x^3}{1 + 3x + 3x^2 + x^3} \right) \ln x = \begin{cases} 0 & \text{if } a=1 \\ \pm \infty & \text{if } a \neq 1 \end{cases}$$

Q10. $\int_2^3 \frac{\ln(x-2)}{x^2} dx$

parts $u = \ln(x-2) \quad v' = \frac{1}{x^2}$
 $u' = \frac{1}{x-2} \quad v = -\frac{1}{x-2}$

$$= \lim_{k \rightarrow 2} \int_k^3 \frac{\ln(x-2)}{x^2} dx$$

$$= \lim_{k \rightarrow 2} \left(\left[-\frac{1}{x} \ln(x-2) \right]_k^3 + \int_k^3 \frac{1}{x-2} \times \frac{1}{x} dx \right)$$

↓ parts of $\frac{A}{x} + \frac{B}{x-2}$

$$\frac{1}{x(x-2)} = \frac{A}{x} + \frac{B}{x-2} \quad A+B=0$$

$$-2A=1 \quad A=-\frac{1}{2} \quad B=\frac{1}{2}$$

$$= \lim_{k \rightarrow 2} \left(\left[-\frac{1}{x} \ln(x-2) \right]_k^3 + \int_k^3 \frac{-\frac{1}{2}}{x} dx + \int_k^3 \frac{\frac{1}{2}}{x-2} dx \right)$$

$$= \lim_{k \rightarrow 2} \left(\left[-\frac{1}{x} \ln(x-2) - \frac{1}{2} \ln x + \frac{1}{2} \ln(x-2) \right]_k^3 \right)$$

$$= \lim_{k \rightarrow 2} \left(0 - \frac{1}{2} \ln 3 + 0 - \left(-\frac{1}{k} \ln(k-2) - \frac{1}{2} \ln k + \frac{1}{2} \ln(k-2) \right) \right)$$

$$= \lim_{k \rightarrow 2} \left(-\frac{1}{2} \ln 3 + \frac{1}{2} \ln k + \left(\frac{1}{k} - \frac{1}{2} \right) \ln(k-2) \right)$$

$$= \lim_{k \rightarrow 2} \left(-\frac{1}{2} \ln 3 + \frac{1}{2} \ln k + \left(\frac{2-k}{2k} \right) \ln(k-2) \right)$$

$$= \lim_{k \rightarrow 2} \left(-\frac{1}{2} \ln 3 + \frac{1}{2} \ln k - \frac{(k-2) \ln(k-2)}{2k} \right)$$

$$= -\frac{1}{2} \ln 3 + \frac{1}{2} \ln 2 = \frac{0}{2k}$$

$$= \frac{1}{2} (\ln 2 - \ln 3)$$

$$= \frac{1}{2} \ln\left(\frac{2}{3}\right)$$

as $\lim_{k \rightarrow 2} (k-2) \ln(k-2)$

$= \lim_{k \rightarrow 0} k \ln k$

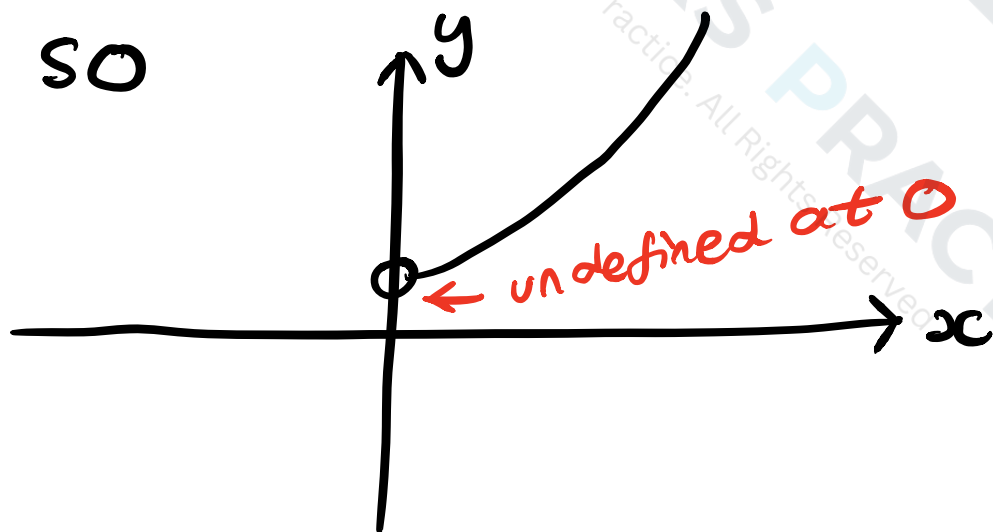
Q11. $f(x) = x^x$

a) As $e^{\ln f(x)} = f(x)$

we have $f(x) = e^{\ln x^x} = e^{x \ln x}$

b) $x \ln x$ is undefined for $x \leq 0$

so



c) $\lim_{x \rightarrow 0} x^x = \lim_{x \rightarrow 0} e^{x \ln x} = \lim_{y \rightarrow 0} e^y = 1$

If $y = x \ln x$ as $x \rightarrow 0$ $y \rightarrow 0$